

CARLOS AGUILAR

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EDUCATION

Hunter College (CUNY) — B.A. Computer Science, Minor in Mathematics *Expected May 2027*
GPA: 3.3/4.0 · **Relevant Coursework:** Software Analysis & Design I–III, Computer Architecture, Data Science, Discrete Mathematics, Calculus I & II, Linear Algebra, Applied Statistics

TECHNICAL SKILLS

Languages: Python, C++, Swift, JavaScript, TypeScript, HTML, CSS
Frameworks & Libraries: React, FastAPI, Pandas, NumPy, Matplotlib, Scikit-Learn
Tools: Git, GitHub, VS Code, Xcode, Jupyter Notebook, SQLite, Linux, OpenGL, SDL2
Concepts: Data Structures, Algorithms, OOP, Statistical Analysis, Software Design, Graphics Programming

PROJECTS

3D Graphics Engine | C++, OpenGL, SDL2, GLSL

- Built a real-time 3D graphics engine from scratch across 6 iterative milestones, culminating in an OpenGL 4.1 renderer with Phong lighting, a first-person camera, and texture mapping.
- Implemented custom data structures: linked list transform pipeline, array-bag rotation registry, and a BST spatial index for efficient 3D point queries.
- Applied inheritance, polymorphism, and abstract classes across Vector3D math, Euler/Quaternion/Axis-Angle rotation systems, and translation/scale transformations.
- Loaded and rendered arbitrary .obj mesh files with per-vertex normals and UV coordinates for texture mapping.

NYC Cafe Analysis | Python, FastAPI, SQLite, React, Pandas

- Built a full-stack data analytics dashboard analyzing 149,000+ NYC coffee shop transactions to surface business insights for cafe owners and customers.
- Designed a SQLite schema with indexed columns and wrote optimized SQL queries for revenue trends, peak hours, top products, and location comparisons.
- Developed a FastAPI backend with 10 REST endpoints supporting location and time-of-day filtering, consumed by a React dashboard with live charts.
- Built a customer-facing view with an interactive time slider showing what products are most popular at any location and hour.

AI Organizer | Swift, Python

- Developing an AI-powered iOS productivity app focused on intelligent task and information organization using SwiftUI and Python backend services.
- Applying software design patterns, testing, and iterative development practices across the full mobile stack.

EXPERIENCE

Programming Tutor · Independent *2023 — Present*

- Assisted students with programming concepts, and algorithm design across multiple languages.

Mathematics Tutor · Independent *2023 — Present*

- Tutored students in algebra, calculus, linear algebra, and statistics; adapted explanations to individual learning styles.

English Language Tutor · Independent *2022 — Present*

- Helped students improve reading comprehension, academic writing, and verbal communication skills.

HONORS & ACHIEVEMENTS

- College Honors / Principal's Honor Roll — multiple consecutive years
- Graduated high school one year early
- GPA: 3.3/4.0 — Hunter College CUNY

LEADERSHIP & ACTIVITIES

- Member, Computer Science Club — Hunter College
- Member, Korean Student Association — Hunter College
- Interests: Hackathons, software engineering, AI/ML, data science, quantitative finance, iOS development